

Emotional and Energy Variations in University Personnel. A VR Intervention

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Figure 1: A sample of the experiences to measure emotional and energy variations. From left to right: a) throwing leaves in a river, b) a fireplace in the night, c) hitting objects with a bat, and d) bursting a balloon.

ABSTRACT

This work expands the knowledge about Virtual Reality (VR) based therapeutic interventions for mental health issues, with a new pilot study directed to university personnel. We use VR applications to promote emotional well-being in university environments, analyzing their influence on emotional management and the energetic state of participants. Our methodology was co-designed with psychologists of our Occupational Health and Safety Department as an alternative for treatment for all personnel. Results show a positive impact on participants' energy levels and mood states, and the need for different approaches for different mood states.

Index Terms: virtual reality, mental health, psychology, university personnel

1 INTRODUCTION

Mental health is a key component in human well-being. Unfortunately, mental health issues are proliferating around the world. Among young people, suicide is now in the top three causes of death for people between 15 and 29 years old [19]. In particular, University students seem especially vulnerable, given the high levels of stress, anxiety, and fatigue they experience due to their workload [17]. This work presents a thorough methodology to incorporate VR experiences as one more alternative of treatment in a University environment. Key elements in our method are a co-design strategy with psychologists and novel ways to measure success of the intervention.

We contribute the analysis based on energy levels and mood states, as metrics that could be interesting for future studies in this topic; the analysis of particular cases and how they can be explained

by previous research; and the description of our intervention in Colombia, a different region of the world than previous studies.

2 RELATED WORK

The emotional well-being subjectively measures the mental balance, related to positive emotions, satisfaction with life, and capabilities to regulate emotions during stressful situations [11]. Emotional regulation is the ability to adequately respond to intense emotional experiences by means of healthy strategies, instead of using disfunctional mechanisms [13].

An energetic state is the subjective measure of vitality, of feeling oneself alive. This measure is conditioned by several factors, both physical (i.e. sleep, exercise, nutrition), and psychological (i.e. positive emotions, life's purpose, positive perspectives, social attachments). The energetic state promotes agency and resiliency, and it is an indication of mental and physical well-being [14], although there is no consensus on the matter, it is assumed that a high energetic state promotes attention, active participation, and learning.

VR has shown benefits related to well-being and mental health. A systematic review shows that VR-based relaxation techniques can be effective to reduce stress [12]. A VR-based intervention in a University shows positive results not only in emotional state, but also in reducing anxiety and sadness [6]. Similarly, [3] shows that VR-based meditation produces better results than conventional methods such as video. In this last study, VR-based meditation before an exam produced better scores and less anxiety. VR beaches or forests have been shown as *virtual havens*, which can reduce anxiety [4]. These treatments complement traditional methods in therapy and they can improve motivation and adherence in patients [2, 7, 9]. Techniques such as the Mindfulness-Based Stress Reduction (MBSR) can reduce levels of anxiety, depression, and stress, by means of techniques such as conscious breathing, meditation, and gentle yoga [10].

3 CO-DESIGN PROCEDURE

We first confirmed a common understanding between psychologists and researchers of the previous work in the VR field. We analyzed

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the needs in our environment and we defined the rules of our collaboration. Second, we defined the procedure with subjects and we submitted our proposal to the ethical committee in our University. We also identified the applications that were part of our pilot study, three from the Meta store and two from our own development. Finally, we gathered the information from subjects and produced the results reported here.

3.1 Agreements and Understanding

Some psychologists in our University had experience with VR technologies and they wanted to offer our students and community in general novel VR applications for treatment of mental health issues. They had seen several applications in the Meta Store, but they wanted a particular development for our University, in order to control better the experience. They had limited budget for initial investment, but they were willing to explore the possibilities that the facilities in our campus could offer.

We performed one demo session for psychologists where we showed a sample of immersive applications in HMDs, especially commercial applications related to the fields they wanted to include. We also had two more meetings where we exposed the limits of our development capabilities and the research questions we wanted to address. After some discussion, we came up with the experimental design described later in this paper.

For hardware, we decided to use the equipment available in our lab. We also considered to use headsets available at our Library, although they required a non-supervised method of use.

3.2 Experimental Design

We used both our developments and commercial applications in order to study advantages and shortcomings of both alternatives. We selected experiences between 5 and 15 minutes in length. We wanted to include subjects with low English proficiency, therefore we selected applications where spoken English was not the main stimuli.

We used Meta Quest 3 HMDs, due to their availability in our lab and other facilities in our University, such as our Library. From the requirements of our psychologists, we included three very different commercial applications: FitXR, where exercise could be an emotional regulator [5, 15]; Nature Treks VR, for the reported benefits of nature in well-being [4]; and Headspace XR, to study the benefits of mindfulness [10, 16].

We decided to develop two applications with opposite stimuli, where we could have more control and log more information about their use. One concentrated on mindfulness and nature contemplation, whereas the other looked for more corporeal movements and psycho-physical activation. The former is called Night Forest and the latter Mood Room. In these applications we capture the time in each task, time and object identifier for each interaction, and the focus of attention. These data is captured in real time and saved in the device in an encrypted file. After one or more participants, we can upload these files to our analysis system.

We looked for volunteers in two Institutions to use these five applications in our lab, at most one per day per participant. Recruitment was performed through social networks in the Institutions and direct invitation from psychologists involved in our pilot study. The only requirement for subjects was their willingness to participate. Per request of our psychologist, in this study we are unaware of their clinical diagnosis at the moment of the test, in order to concentrate in the pilot nature of the study. First, subjects had to read and fill an informed consent form, that included information about the risks of VR for people with a history of epilepsy, vestibular anomalies, motion sickness, diagnosed psychological conditions or severed visual impairments. Later, each subject had to fill one survey before the experience in order to measure its initial mental state;

and another survey after the experience in order to measure the effect of a particular application. Apart from the data collected in our applications, we asked subjects to register their mood and their perceived energy state. Each intervention was designed to last 30 minutes: 5 min for subject reception, 5 min for the initial questionnaire, 5–10 min for the experience, and 10 min for the final questionnaire and closure.

The reported demographic data includes age, genre, role within the University (student, professor, administrative, personal from the Office of Health, personal from General Services, Other), pre-existing medical conditions (epilepsy, vertigo, visual impairment, photosensitivity), techniques for emotional well-being that use frequently (meditation, deep breathing, mindfulness, yoga), and previous experience with VR equipment.

The pre-intervention questionnaire asked subjects to identify the main feeling they had at that moment from the following 15 options, suggested by the psychologists in the team and including the five basic ones identified by [1]: happy, sad, anger, afraid, disgust, satisfaction, boredom, frustration, shame, contempt, peace, nostalgia, hate, anxiety, repulsion. For the selected emotion, participants had to score their degree from a scale from 0 (very low) to 10 (very intense). We also used a scale from 0 to 10 to score the state of energy that subjects perceived from themselves.

The post-intervention questionnaire included the same questions than the previous one for the state of emotion and energy. We also asked subjects if they believe the experience was good for them and how they describe such experience for their well-being.

3.3 VR Software

Both applications were developed in Unity v. 2022.3.38f1, with the Meta XR All-in-One SDK. Both applications share the components for navigation, interaction, setup, and encryption of personal data. Source code and extra material is available at [8].

Night Forest used the LowPoly Forest - Lite package from the Meta Store for its appearance. It shows a night scene in a natural scenario, as shown in Figure 2. Subjects could choose between four experiences in any order. An interactive tree allowed subjects to take leaves and throw them to a river (Figure 1.a). A second experience started a fireplace once four stones were collected (Figure 1.b). A third experience invited subjects to synchronize their breathing with the expansion and contraction of a sphere. Finally, a mystery box showed an animation of fireflies once it was opened. All experiences included a voice narrating a mindfulness-related script. For example, the audio in the interactive tree said *Take a leave with your fingers. Move it softly while you load in it all emotions you want to set free. When you release it, observe how the river takes it away, with the emotions you liberated.* At the end of the experience subjects were invited to add their id number, which is used to cross-reference the data captured in the application with the data from questionnaires.

Mood Room was designed to promote vigorous movements from subjects. At the beginning, the scene has red tones, as it can be seen from the last two images in Figure 1. The entire scene transitions to a white tone after interaction (Figure 3), showing with colors an emotional transition. It uses objects and a scene from the Meta XR All-in-One SDK. It shows three active experiences of two minutes each. In the first experience participants collect points by hitting objects in the room with a bat (Figure 1.c). The second experience shows five red buttons and users are invited to press the ones that change color to green as fast as they can. Finally, users are invited to aim to balloons and burst them (Figure 1.d). All tasks are accompanied by a voice that explains their purpose, and points are collected to motivate users with a final score. At the end of the three experiences, a dialog is shown to collect the user's id, so recorded events could be correlated with the information in questionnaires.



Figure 2: Night Forest, an application for relaxation.



Figure 3: Mood Room, an application to relieve agitated emotions.

3.4 Data Gathering and Analysis

We collected complete data from 41 participants, between 18 and 45 years old. Their average age is 26 years and median 22. 53.7% are undergrad students, followed by 22% of administrative personnel and 14.6% from the Occupational Health and Safety Department (others are mostly graduate students). They used our system 65 times: 2 users used 4 applications, 4 users used 3, 10 users 2, and 25 just one application. The most used app was FitXR (38.5%), followed by Nature Treks (21.5%), by both Mood Room and Night Forest (15.4%), and by Headspace XR at last (9.2%).

4 SELECTED RESULTS

Subjects declared great changes in emotions after the intervention. Figure 4 shows the declared emotions before the intervention, for all VR experiences. We notice that the main emotions reported are anxiety, frustration, boredom, and happiness.

Figure 5 shows the declared emotions after the intervention. In this case, the main emotions were peace, satisfaction, happiness, and nostalgia.

These changes are also noticeable in each one of the applications, with different emotions. If we concentrate in the more active apps, Figure 6 shows the changes in emotions before and after the intervention with FitXR. The four most common emotions before the intervention are anxiety, frustration, satisfaction, and happiness. After the intervention subjects changed to satisfaction, happiness, peace, and frustration.

Figure 7 shows the reported emotions before and after the intervention with Mood Room, one of our developed applications. In this case, subjects reported anxiety, frustration, happiness, and boredom before the intervention. After the intervention these emotions changed to satisfaction, happiness, peace, and boredom.

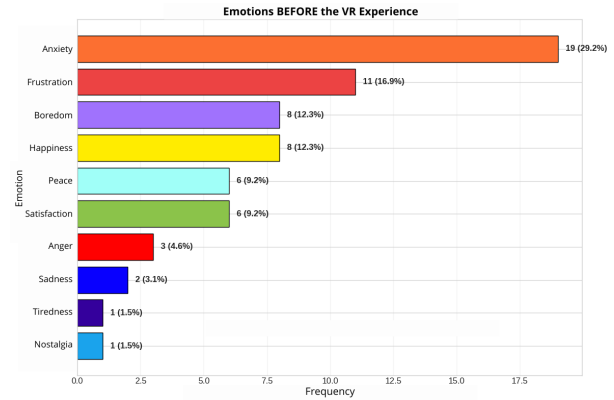


Figure 4: Declared emotions before the experience.

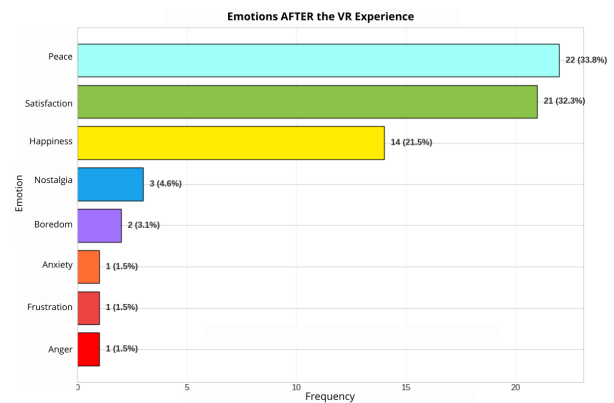


Figure 5: Declared emotions after the experience.

We asked subjects to report the level of intensity of their emotions before and after the experiment. First, there were few individuals who reported the same feeling before and after the intervention. Figure 8 shows how the intensity of their emotions changed, subtracting the intensity after the intervention from the intensity before. We can notice a positive effect in the intensity of satisfaction, peace, and happiness, whereas a negative effect (i.e. reduced intensity) for anger, anxiety, boredom, and frustration.

We can also observe the distribution of intensities per emotion before and after the experiment, for some of the emotions reported. Figures 9, 10, and 11 show the distributions of intensities for subjects that report respectively satisfaction, happiness, and boredom before and after the experiment. We can notice higher levels of intensity in emotions such as satisfaction and happiness, whereas less people and less intensity in subjects with boredom.

4.1 Discussion

First, the changes in emotion within participants are promising. 63% of participants declared negative emotions such as anxiety, frustration, boredom, or anger at the beginning of our experiment. At the end of the experiment, only 7.6% of participants declared one of such emotions. Similarly, only 30.7% of participants declared at the beginning of the experiment a positive emotion such as happiness, peace, or satisfaction. That percentage grew to 87.6% at the end of the experiment.

Applications that promote activity such as FitXR and Mood Room showed more participants with satisfaction and happiness at



Figure 6: Declared emotions before and after the intervention with FitXR.

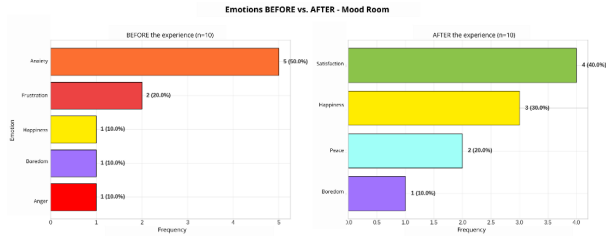


Figure 7: Declared emotions before and after the intervention with Mood Room.

the end of the intervention (Figures 6 and 7), whereas applications inspired in mindfulness such as Nature Treks and Night Forest resulted mostly in peace and nostalgia. We believe this shows the importance of different approaches for VR based treatment, where many results are possible. A psychologist also emphasized the importance of nostalgia after some mindfulness-related experiences. Although it is not possible from the data to identify if this is a positive or negative effect, it has been reported as a normal response to loneliness or after a personal loss [18].

From the particular emotional transitions in general, we can notice that participants started from negative emotions such as anxiety, frustration, or anger and ended up in positive emotions such as peace, satisfaction, or happiness. A psychologist also considered this change as promising, due to the nature of the emotions before and after the experiment. There are also interesting changes in the intensity and level of energy. The intensity of emotions usually decreased or remained the same in subjects starting in negative emotions. At the same time, energy levels mostly increased, which

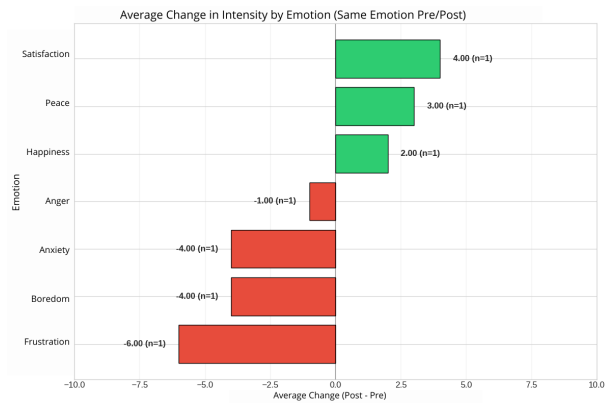


Figure 8: Changes in intensity (post-pre), subjects with the same emotion.

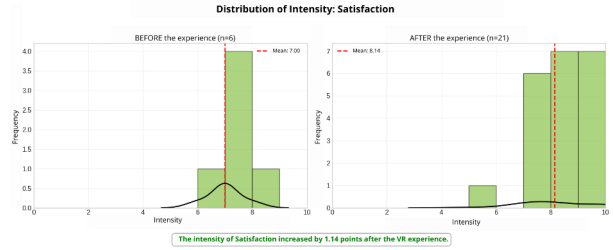


Figure 9: Distribution of intensities from subjects with satisfaction, before and after the experiment.

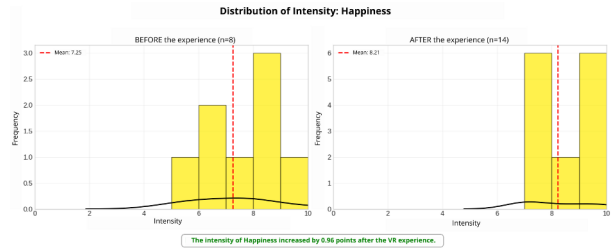


Figure 10: Distribution of intensities from subjects with happiness, before and after the experiment.

could be an indicator of well-being improvement.

An analysis of individual cases gives also interesting insights. Anxiety for one user of Night Forest was reduced by 4. Another user reduced his or her anger by 1 in the same application. Just another user reduced his or her frustration from 9 to 2 with FitXR. While individuals experience mood changes in different ways and with different intensities depending on several factors in their environment, subjects in our study generally reduced the intensity of their negative emotions.

Since volunteers did not attend to all the programmed sessions, the analysis is limited to descriptive statistics. Nevertheless, these findings show us key directions for future, more rigorous studies. Our data suggest that these types of applications can produce different but complementary results in the well-being of participants. Therefore, it is important to offer different and complementary experiences to people so we can attend better their particular well-being needs. The potential of these applications to positively influence people's energy levels and emotional intensity is evident. However, a study with a larger population and even samples in all conditions will be key to better understand how interactions within these virtual environments affect the behavior and emotions of people.

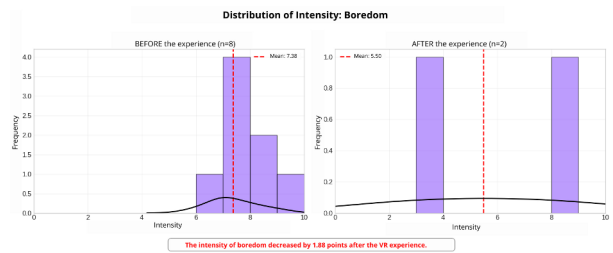


Figure 11: Distribution of intensities from subjects with boredom, before and after the experiment.

5 LIMITATIONS OF OUR WORK

There are several limitations in our work that might limit the scope of our results. First, participants were recruited, independently of their need for treatment. We did not include a control group, due to restrictions in time and number of volunteers. A more focused and complete sample could show different results. However, we believe our results show promising positive benefits in a general population within a University, a population that includes the ones needing special treatment.

The pilot nature of our study has also limitations, in terms of the number of interventions that each participant experienced. Future studies could study the lasting effects of VR treatment over a longer period of time and with more even samples between conditions.

Our sample is also limited. Future studies with more participants could show a more clear picture of the benefits of VR in this context.

6 CONCLUSIONS

We have presented a pilot study on the benefits of VR interventions for University personnel. Our report shows how participants obtained benefits from the experiences, in terms of more positive emotions after the intervention and more energy. We also analyzed the intensity of emotions and the energetic state of participants, as promising metrics in this field.

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